MCM Research Notes

Reinterpreting Physics Through the Monistic Continuum Model

No. 1

Physical Constants as Observational Quantities

Measurement versus Cosmological Extrapolation

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The MCM Research Notes are an ongoing publication series dedicated to the reinterpretation of established physical concepts within the framework of the Monistic Continuum Model (MCM 10.1).
The objective of the series is not to refute established physics, but to investigate possible deeper interpretations of experimentally confirmed observations.

Preface to the Series

The *MCM Research Notes* pursue an approach different from that of many alternative physical models.

Their objective is not to replace established physics, but to examine well-known physical observations from a new perspective.

The Monistic Continuum Model Theory (MCM 10.1) accepts all experimentally confirmed results of modern physics.

In particular, the experimental achievements of

- Relativity Theory,
- Quantum Theory,
- Cosmology,
- Particle Physics

are not disputed.

The central question is:

> How can established physical observations be reinterpreted within a dynamic continuum?

Each issue addresses a clearly defined question.

The series follows four guiding principles:

1. Acceptance of all experimentally confirmed observations.
2. Strict separation between measurement and interpretation.
3. Mathematical and conceptual consistency.
4. Physical plausibility instead of speculative claims.

This first issue examines the role of physical constants and investigates the relationship between observed constancy and cosmological extrapolation.

Abstract

Modern physics is based upon a number of fundamental physical constants, including the speed of light, Planck's constant, and the gravitational constant.

These quantities are measured with high precision and are observed to be constant.

The Monistic Continuum Model Theory (MCM 10.1) fully accepts these observations.

At the same time, it distinguishes between the directly observed constancy of a physical quantity and claims regarding its validity throughout the entire cosmological past and future.

This issue formulates that distinction as a methodological principle and examines the question of why constant natural quantities are observed within a dynamic cosmos.

MCM treats the stability of physical constants not as a self-evident assumption, but as an independent physical problem requiring explanation.

Keywords

Monistic Continuum Model,
MCM 10.1,
Physical Constants,
Speed of Light,
Planck Constant,
Cosmology,
Philosophy of Physics,
Observation,
Extrapolation,
Dynamic Cosmos

1 Introduction

Modern physics relies upon a set of fundamental constants.

Among the most important are

$$c = 299\,792\,458 \text{ m/s},$$

$$h = 6.62607015 \times 10^{-34} \text{ Js},$$

and

$$G = 6.67430 \times 10^{-11} \text{ m}^3\,\text{kg}^{-1}\,\text{s}^{-2}.$$

These quantities form the foundation of Relativity Theory, Quantum Theory, and large parts of modern Cosmology.

MCM 10.1 does not dispute any of these measurements.

The question addressed in this issue is therefore not

Have the physical constants been measured correctly?

but rather

Which statements arise directly from observation and which statements arise only through cosmological interpretation?

The answer to this question provides the foundation for all future MCM Research Notes.

2 The Established Physical View

Modern physics treats physical constants as universal constants of nature.

The speed of light appears, for example, in

$$E = mc^2,$$

while Planck's constant appears in

$$E = hf.$$

The associated theories belong to the most successful scientific models in human history.

MCM 10.1 explicitly acknowledges these achievements.

Nothing in this issue questions the experimental validity of these results.

3 The MCM Position

MCM adopts a deliberately cautious formulation.

The following statements hold:

$$\boxed{c := \text{constant}}$$

$$\boxed{h := \text{constant}}$$

$$\boxed{G := \text{constant.}}$$

The notation

$$:=$$

is crucial.

It denotes an observed fact.

It does not constitute a statement about the entire past or future of the Universe.

4 Measurement and Extrapolation

MCM distinguishes between two types of scientific statements.

4.1 Measurement

A measurement is a directly observable statement.

For example, the present measurement is

$$c = 299\,792\,458 \text{ m/s}.$$

This statement is fully accepted.

4.2 Extrapolation

An extrapolation extends the range of a measurement beyond direct observation.

For example:

- The speed of light had exactly the same value one hundred million years ago.
- The speed of light will have exactly the same value one hundred million years in the future.

Such statements may be justified within a theoretical framework.

However, they are not direct measurements.

5 The Problem of the Cosmological Epoch

At this point MCM 10.1 introduces a further distinction.

Every measurement is obtained within a specific stage of cosmic evolution.

Today's measurements were obtained neither in the early Universe nor in the distant future.

They were obtained during the present cosmological epoch.

MCM therefore asks:

> What influence does the cosmological epoch have on the interpretation of a measurement?

The statement

$$\boxed{c_{\text{today}} = 299\,792\,458 \text{ m/s}}$$

is experimentally established.
However, it does not automatically imply

$$\boxed{c_{\text{past}} = c_{\text{today}} = c_{\text{future}}.}$$

The first statement is an observation.
The second statement is a cosmological inference.
MCM therefore distinguishes between

observed constancy

and

cosmological invariance.

Formally,

$$\boxed{\text{Observed Constancy} \neq \text{Cosmological Invariance.}}$$

This does not imply that MCM claims physical constants vary.
The statement merely expresses that present-day observations do not automatically prove the unlimited validity of the same values throughout the entire history of the Universe.

> Present observations do not automatically establish the eternal validity of the same quantities throughout the entire history of the Universe.

6 The Dynamics Problem of Physical Constants

Modern cosmology describes a cosmos undergoing continuous evolution.
Observed processes include:

- the expansion of the Universe,
- the formation and evolution of stars,
- the formation and evolution of galaxies,
- the development of large-scale structures,
- the formation and decay of astrophysical systems.

In other words:

$$\boxed{\text{The Cosmos is Dynamic.}}$$

At the same time, physical constants such as

$$c, \quad h, \quad G$$

are treated as invariant.
For established physics this is not regarded as a contradiction, because these quantities are considered fundamental properties of nature.
At this point MCM raises an additional question:

Why do physical constants remain unaffected
by cosmological dynamics?

This question should not be interpreted as a claim that physical constants vary.
MCM neither claims

$$c \neq \text{constant}$$

nor

$$h \neq \text{constant}.$$

Instead, it asks:

How can the observation of constant physical quantities
be explained within a dynamic cosmos?

From the perspective of MCM, the constancy of physical constants is itself a phenomenon requiring explanation.

If cosmic evolution is real, then the observation of invariant natural quantities does not appear self-evident, but rather as a property of nature that itself deserves explanation.

This leads to the central question:

Is the constancy of physical constants fundamental
or itself the result of a
deeper physical process?

MCM does not answer this question in the present issue.
However, it identifies it as a fundamental research problem.

This extends the previous distinction between measurement and extrapolation by introducing an additional aspect:

Dynamic Cosmos $\Longrightarrow$ Need to Explain Observed Constancy.

7 The Principle of Epistemological Caution

From these considerations follows a fundamental methodological principle:

Measurement and cosmological extrapolation are not identical.

Observed constancy is accepted.
Cosmological universalization requires additional justification.
MCM therefore makes a deliberate distinction between

experiment

and

interpretation.

8 Compatibility with Established Physics

MCM 10.1 continues to accept all experimentally confirmed relations of modern physics.

In particular, the following remain valid:

$$E = mc^2,$$

$$E = hf,$$

as well as the empirically confirmed predictions of Relativity Theory and Quantum Theory.

MCM does not replace these results.

It adopts them as established observations.

9 The Perspective of MCM

Within MCM, physical constants are initially treated as boundary conditions of observation.

Accordingly,

$$\boxed{c := \text{constant}}$$

$$\boxed{h := \text{constant}}$$

$$\boxed{G := \text{constant.}}$$

The objective of a deeper theory is not to discard these constants.

Its objective is to explain why they appear constant within observable physics.

10 Research Objective

MCM therefore does not seek the elimination of physical constants.

Instead, it asks:

> Why do physical constants appear constant within observable physics?

A successful answer would not replace established physics.

Rather, it would provide a deeper foundation for it.

11 Working Hypothesis of MCM

The provisional working hypothesis of MCM 10.1 is:

> Physical constants are experimental facts.

and

> Statements concerning their cosmological past or future require independent evidence.

12 Testable Consequence

The position formulated in this issue leads to an immediately testable consequence.

MCM distinguishes between

measurement

and

cosmological inference.

From this distinction follows directly:

> The constancy of a physical constant within observable physics must not automatically be interpreted as proof of its eternal constancy.

MCM therefore formulates the following testable requirement:

> Any statement concerning the cosmological past or future of a physical constant must be justified independently of its present-day measurement.

This requirement is further strengthened by the previously formulated Dynamics Problem of Physical Constants.

If the cosmos undergoes a continuous process of development,

> The Cosmos is Dynamic.

then the following question naturally arises:

> Why do physical constants appear constant within a dynamic cosmos?

MCM does not regard this question as solved, but as an independent research problem.

From this follows a concrete evaluation criterion for any physical or cosmological theory:

1. Which physical constants are assumed?
2. Which of these constants have actually been measured?
3. Which statements about the past or future arise only from model assumptions?
4. How is the observed constancy of these quantities physically explained within a dynamic cosmos?

MCM predicts that a substantial portion of cosmological statements belongs to the third category.

This makes a systematic distinction between observation, model assumptions, and cosmological interpretation possible.

The testable consequence of Issue 1 is therefore:

> The further a statement is removed from direct observation, the more strongly it depends upon underlying model assumptions.

MCM expects in particular that statements concerning the entire cosmological past and future depend more strongly upon the assumptions of their underlying models than statements concerning directly observable present-day physics.

This leads to an additional testable question:

> Does a dynamic cosmos necessarily require an explanation for the observed constancy of physical constants?

MCM does not answer this question in the present issue.

However, it requires that every theory of a dynamic cosmos explicitly explain why physical constants may be regarded as invariant and on what basis such a conclusion is reached.

13 Conclusion

MCM does not begin with speculation about varying constants.

It begins with observations.

At the same time, it demands a clear distinction between what has been measured and what has been inferred from those measurements concerning the entire cosmological history of the Universe.

Furthermore, it identifies the observed constancy of physical constants within a dynamic Universe as a scientific problem in its own right.

This issue therefore establishes the methodological foundation of the entire *MCM Research Notes* series:

> Observation takes precedence over extrapolation.

> Interpretations may extend observations, but must not be confused with them.

> The constancy of physical constants is not to be discarded, but to be physically explained.

MCM Research Notes

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The author would like to thank all readers who are willing to examine fundamental questions of physics from new perspectives.

Special thanks are extended to the scientific community whose centuries of work provide the foundation for all observations, experiments, and theories discussed throughout this series.

The *MCM Research Notes* are not intended as a replacement for established physics, but as a contribution to its interpretation and further theoretical development.

Scientific progress emerges through the combination of observation, critical analysis, and open discourse.

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Series Information

MCM Research Notes

Reinterpreting Physics Through the Monistic Continuum Model

The purpose of this series is to analyze established physical observations from the perspective of MCM 10.1.

Each issue addresses a clearly defined question and follows the principles of:

- acceptance of all experimentally confirmed observations,
- distinction between measurement and interpretation,
- search for deeper physical explanations,
- complete traceability and transparency of reasoning.